Asteria University of Engineering and Applied Sciences

BEng programs, curriculum, and graduation requirements

Purpose and award

This fictional catalog is a synthetic test document for CampusGPT. Asteria University of Engineering and Applied Sciences is invented; program names and policies are sample data.

The Bachelor of Engineering (BEng) is a four-year, full-time undergraduate degree organized into eight semesters. Students may complete the program in up to six academic years, subject to approved leave and the university academic regulations.

A BEng is awarded after completion of 160 credits, all required core and program courses, the engineering design sequence, a supervised capstone project, professional practice, and all financial and academic clearance requirements.

Admission and first-year foundation

Applicants normally need a secondary-school completion credential with mathematics, physics, and chemistry or approved equivalents. Admission is competitive and subject to the annual university admissions notice.

All BEng students complete a shared foundation in calculus, linear algebra, physics, chemistry for engineers, programming, engineering graphics, communication, ethics, and introductory design. Departmental specialization begins in the second year.

Credit structure and progression

The 160-credit minimum consists of 32 credits of mathematics and basic sciences; 48 credits of engineering fundamentals and common core; 52 credits of departmental required courses; 16 credits of technical electives; 4 credits of humanities and communication; and 8 credits of design capstone and professional practice.

A typical semester carries 18 to 22 credits. Students must pass prerequisite courses before enrolling in dependent courses. A cumulative grade-point average of at least 2.00 on a 4.00 scale is required for graduation. A department may require a higher average in designated safety-critical courses.

A student who fails a required course must repeat it or complete an approved equivalent. Credit is not awarded twice for substantially equivalent courses.

Eight-semester study plan

Semester 1: Calculus I, Physics for Engineers I, Programming Fundamentals, Engineering Graphics, Communication for Engineers.

Semester 2: Calculus II, Physics for Engineers II, Chemistry for Engineers, Engineering Design Studio I, Ethics and Society.

Semester 3: Differential Equations, Probability and Statistics, Materials Science, Circuits and Systems, Departmental Foundations I.

Semester 4: Numerical Methods, Thermodynamics or Systems Foundations, Engineering Design Studio II, Departmental Foundations II, Technical Writing.

Semester 5: Departmental Core I, Departmental Core II, Measurement and Instrumentation, Technical Elective I, Engineering Economics.

Semester 6: Departmental Core III, Departmental Core IV, Technical Elective II, Laboratory Practice, Research Methods.

Semester 7: Departmental Advanced Core, Technical Elective III, Capstone Design I, Professional Practice or approved internship.

Semester 8: Departmental Advanced Core, Technical Elective IV, Capstone Design II, Project Demonstration, Safety and Professional Responsibility.

BEng departments and sample specialization courses

Civil Engineering: structural analysis, geotechnical engineering, transportation systems, water resources, construction methods, and sustainable infrastructure design.

Environmental Engineering: environmental chemistry, hydrology, water treatment, air quality, waste systems, and environmental impact assessment.

Computer Engineering: digital logic, computer architecture, operating systems, embedded computing, networks, and hardware-software integration.

Software Engineering: data structures, software architecture, databases, testing and quality, secure software, and team software studio.

Electrical Engineering: circuit analysis, electrical machines, power systems, control systems, and renewable energy conversion.

Electronics Engineering: analog and digital electronics, signal processing, communication systems, microcontrollers, and embedded design.

Mechanical Engineering: mechanics of materials, fluid mechanics, heat transfer, machine design, manufacturing, and mechatronics.

Aerospace Engineering: aerodynamics, flight mechanics, propulsion, aerospace structures, and aircraft systems.

Chemical Engineering: transport phenomena, reaction engineering, process control, separation processes, and plant design.

Materials Engineering: physical metallurgy, polymers, ceramics, characterization, and materials selection.

Industrial Engineering: operations research, production systems, quality engineering, logistics, and human factors.

Biomedical Engineering: biomechanics, biomaterials, medical instrumentation, physiological systems, and healthcare technology design.

Capstone, internship, and academic integrity

The capstone is a two-semester team design project. Students define a need, establish requirements, analyze alternatives, build or simulate a solution, evaluate safety and sustainability, and present a final report and demonstration. A faculty supervisor and departmental review panel assess the project.

Professional practice is satisfied by a supervised internship of at least eight weeks, an approved cooperative placement, or a department-approved engineering practice project. Students submit a reflective report and employer or supervisor evaluation.

All work must comply with university academic integrity rules, laboratory safety procedures, accessibility requirements, and applicable engineering codes of practice.

BEng completion checklist

Confirm at least 160 earned credits and all degree-audit categories are satisfied.

Pass every required course and prerequisite, including the mathematics and science foundation.

Complete four technical electives approved for the declared department.

Pass Capstone Design I and II and the public project demonstration.

Complete the approved internship or professional practice alternative.

Maintain a cumulative GPA of at least 2.00 out of 4.00 and resolve incomplete grades.

File the graduation application with the Registrar and receive department, library, and financial clearance.